

## HÖRMANDER'S THEOREM FOR THE CUACHY-RIEMANN OPERATOR: THE ONE VARIABLE CASE

The entire  $L^2$  theory revolves around solving the  $\bar{\partial}$ -problem.  
Recall that

$$\frac{\partial}{\partial z} = \frac{1}{2} \left( \frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right) \text{ and } \frac{\partial}{\partial \bar{z}} = \frac{1}{2} \left( \frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right).$$

We will use the following standard notations throughout the notes  
 $D = \frac{\partial}{\partial z}$  and  $\bar{D} = \frac{\partial}{\partial \bar{z}}$ . We also define the complex Laplacian operator  
as  $\Delta = D\bar{D}$ , which differs from the usual laplacian by a factor of 4.

We will assume that all the functions are complex-valued (unless  
stated otherwise) and measurable.

**Theorem 1.** *Let  $\Omega \subset \mathbb{C}$  be a domain and let  $\varphi \in \mathcal{C}^2(\Omega)$  be a strictly  
subharmonic function, i.e.,  $\varphi$  is real valued and  $\Delta\varphi = D\bar{D}\varphi > 0$ . If  $g$  is  
a function on  $\Omega$  such that  $\int_{\Omega} \frac{|g|^2}{\Delta\varphi} e^{-\varphi} dV < \infty$  then there is a function  $u$   
on  $\Omega$  such that*

$$(1) \quad \frac{\partial u}{\partial \bar{z}} = g,$$

$$(2) \quad \int_{\Omega} |u|^2 e^{-\varphi} dV \leq \int_{\Omega} \frac{|g|^2}{\Delta\varphi} e^{-\varphi} dV.$$

*Remark.* 1. The equality in (1) holds in the sense of distributions,  
(since  $u$  is a measurable function). That is, for every  $\alpha \in \mathcal{C}_c^\infty(\Omega)$ , we  
have

$$\int_{\Omega} -u \overline{\left( \frac{\partial \alpha}{\partial z} \right)} dV = \int_{\Omega} g \bar{\alpha} dV$$

2. This theorem gives a solution of the  $\bar{\partial}$ -problem with estimates on  $u$ .  
This leads to the following corollary.

**Corollary 2.** *Let  $\Omega \subset \subset \mathbb{C}$  be a relatively compact domain in  $\mathbb{C}$ . Then  
there exists a constant  $C_{\Omega}$  such that for each  $g \in L^2(\Omega)$  there is a function  
 $u \in L^2(\Omega)$  such that  $\frac{\partial u}{\partial \bar{z}} = g$  and  $\|u\|_{L^2(\Omega)} \leq C_{\Omega} \|g\|_{L^2(\Omega)}$ .*

*Proof.* Let  $\varphi(z) = |z|^2$ . Since  $\Omega$  is bounded, we can choose  $R > 0$  large  
enough so that  $\Omega \subset D(0; R)$ . This means that for every  $z \in \Omega$

$$e^{-R^2} \leq e^{-|z|^2} \leq 1$$

and since  $g \in L^2(\Omega)$ , the last inequality gives

$$\int_{\Omega} \frac{|g|^2}{\Delta\varphi} e^{-\varphi} dV = \int_{\Omega} |g|^2 e^{-|z|^2} dV < \int_{\Omega} |g|^2 dV < \infty.$$

This shows that  $g$  and  $\varphi(z) = |z|^2$  meet the hypotheses of Theorem 1. ( $\varphi$  is a strictly subharmonic function with  $\Delta\varphi = 1$ ) Thus, by Theorem 1 there exists a  $u$  which is the solution to the  $\bar{\partial}$  equation, and the estimate in (2) gives

$$e^{-R^2} \int_{\Omega} |u|^2 dV \leq \int_{\Omega} |u|^2 e^{-\varphi} dV \leq \int_{\Omega} \frac{|g|^2}{\Delta\varphi} e^{-\varphi} dV \leq \int_{\Omega} |g|^2 dV.$$

This gives  $C_{\Omega} = e^{R^2/2}$ . ■

Note that this is not the best estimate of  $C_{\Omega}$ . We can improve this estimate a little by translating our domain so that the origin is contained in it and then we can have  $C_{\Omega} = e^{(d(\Omega)/2)^2}$ , where  $d(\Omega)$  is the diameter of  $\Omega$ . But the best estimate of  $C_{\Omega}$  is of the order of  $d(\Omega)$ .

*Remark.* Turns out that this constant  $C_{\Omega}$  is the highest eigenvalue of the complex Laplacian.

Let us now look at the proof of the one variable Hörmander's theorem. We break down the proof into three steps.

Step 1: Reformulation of the problem in  $L^2(\Omega, e^{-\varphi})$  space.

Step 2: Computation of the formal adjoint of  $\bar{D}$  in  $L^2(\Omega, e^{-\varphi})$ .

Step 3: Obtain apriori estimates: For each  $\alpha \in \mathcal{C}_c^2(\Omega)$ ,

$$\left\| \bar{D}_{\varphi}^* \alpha \right\|_{\varphi}^2 \geq \langle \Delta\varphi \alpha, \alpha \rangle_{\varphi}, \text{ i.e.}$$

$$\int_{\Omega} \left| \bar{D}_{\varphi}^* \alpha \right|^2 e^{-\varphi} dV \geq \int_{\Omega} \Delta\varphi |\alpha|^2 e^{-\varphi} dV$$

*Proof of Theorem 1. Step 1:* We consider the weighted  $L^2$  space

$$L^2(\Omega, e^{-\varphi}) = \left\{ f : \Omega \rightarrow \mathbb{C} : \int_{\Omega} |f|^2 e^{-\varphi} dV < \infty \right\}$$

with the natural inner product on it

$$\langle f, g \rangle = \int_{\Omega} f \bar{g} e^{-\varphi} dV.$$

**Step 2:** We define the formal adjoint of  $\bar{D}$  in  $L^2(\Omega, e^{-\varphi})$  as: For every  $u \in L^2(\Omega, e^{-\varphi})$  and for every  $\alpha \in \mathcal{C}_c^2(\Omega)$

$$\langle \bar{D}u, \alpha \rangle_{\varphi} = \langle u, \bar{D}_{\varphi}^* \alpha \rangle_{\varphi},$$

if  $\overline{D}u \in L^2(\Omega, e^{-\varphi})$ . Let us now compute the formal adjoint of  $\overline{D}$  in  $L^2(\Omega, e^{-\varphi})$ . Let us first look at the simple case when  $\varphi = 0$ . Here  $u \in L^2(\Omega)$ ,  $Du \in L^2(\Omega)$ , and  $v \in \mathcal{C}_c^2(\Omega)$ . So by the definition of formal adjoint we have

$$\langle \overline{D}u, v \rangle = \langle u, \overline{D}^* v \rangle.$$

But

$$\langle \overline{D}u, v \rangle = \int_{\Omega} \frac{\partial u}{\partial \bar{z}} \bar{v} dV = \int_{\Omega} \frac{1}{2} \left( \frac{\partial u}{\partial x} + i \frac{\partial u}{\partial y} \right) \bar{v} dV.$$

Using integration by parts, we have

$$\int_{\Omega} \frac{\partial u}{\partial x} \bar{v} dV = \int_{\Omega} \frac{\partial(u\bar{v})}{\partial x} dV - \int_{\Omega} u \frac{\partial(\bar{v})}{\partial x} dV.$$

Green's theorem gives us

$$\int_{\Omega} \frac{\partial(u\bar{v})}{\partial x} dx dy = \int_{\partial\Omega} u \bar{v} dy = 0 \text{ (since } v = 0 \text{ on } \partial\Omega).$$

So,

$$\int_{\Omega} \frac{\partial u}{\partial x} \bar{v} dV = - \int_{\Omega} u \frac{\partial(\bar{v})}{\partial x} dV.$$

This shows that

$$\left( \frac{\partial}{\partial x} \right)^* = - \frac{\partial}{\partial x}.$$

Using similar calculations, one can check that

$$\left( i \frac{\partial}{\partial y} \right)^* = -i \frac{\partial}{\partial y}.$$

Using these two results, we have

$$\begin{aligned} \langle \overline{D}u, v \rangle &= \int_{\Omega} \frac{1}{2} \left( \frac{\partial u}{\partial x} + i \frac{\partial u}{\partial y} \right) \bar{v} dV \\ &= - \int_{\Omega} \frac{1}{2} u \left( \frac{\partial \bar{v}}{\partial x} + i \frac{\partial \bar{v}}{\partial y} \right) dV \\ &= - \int_{\Omega} u \frac{\partial \bar{v}}{\partial \bar{z}} dV \\ &= - \int_{\Omega} u \overline{\left( \frac{\partial v}{\partial z} \right)} dV \\ &= \langle u, -Dv \rangle \end{aligned}$$

So when  $\varphi = 0$ ,  $\overline{D}^* = -D$ . Now let us look at the case when  $\varphi \neq 0$ . Since  $v \in \mathcal{C}_c^2(\Omega)$ ,  $ve^{-\varphi} \in \mathcal{C}_c^2(\Omega)$ . So using above result we have

$$\begin{aligned} \int_{\Omega} \frac{\partial u}{\partial \bar{z}} ve^{-\varphi} dV &= - \int_{\Omega} u \overline{\left( \frac{\partial}{\partial z} (ve^{-\varphi}) \right)} dV \\ &= - \int_{\Omega} u \overline{\left( \frac{\partial v}{\partial z} \right)} e^{-\varphi} dV + \int_{\Omega} u \bar{v} e^{-\varphi} \frac{\partial \varphi}{\partial \bar{z}} dV \\ &= \int_{\Omega} u \overline{\left( -\frac{\partial v}{\partial z} + \frac{\partial \varphi}{\partial z} v \right)} e^{-\varphi} dV. \end{aligned}$$

Hence  $\overline{D}_{\varphi}^* = -D + \varphi_z$ .

**Step 3:** Note that

$$\begin{aligned} [\overline{D}, \overline{D}_{\varphi}^*] \alpha &= \overline{D} \overline{D}_{\varphi}^* \alpha - \overline{D}_{\varphi}^* \overline{D} \alpha \\ &= \overline{D} (-D\alpha + \varphi_z \alpha) - (-D\overline{D}\alpha + \varphi_z \overline{D}\alpha) \\ &= -\Delta \alpha + \Delta \varphi \alpha + \varphi_z \overline{D}\alpha + \Delta \alpha - \varphi_z \overline{D}\alpha \\ &= \Delta \varphi \alpha. \end{aligned}$$

So  $[\overline{D}, \overline{D}_{\varphi}^*] = \Delta \varphi$ . To get the apriori estimate,

$$\begin{aligned} \langle \Delta \varphi \alpha, \alpha \rangle_{\varphi} &= \left\langle [\overline{D}, \overline{D}_{\varphi}^*] \alpha, \alpha \right\rangle_{\varphi} \\ &= \left\langle \overline{D} \overline{D}_{\varphi}^* \alpha - \overline{D}_{\varphi}^* \overline{D} \alpha, \alpha \right\rangle_{\varphi} \\ &= \left\langle \overline{D} \overline{D}_{\varphi}^* \alpha, \alpha \right\rangle_{\varphi} - \left\langle \overline{D}_{\varphi}^* \overline{D} \alpha, \alpha \right\rangle_{\varphi} \\ &= \left\langle \overline{D}_{\varphi}^* \alpha, \overline{D}_{\varphi}^* \alpha \right\rangle_{\varphi} - \left\langle \overline{D} \alpha, \overline{D} \alpha \right\rangle_{\varphi} \\ &= \left\| \overline{D}_{\varphi}^* \alpha \right\|_{\varphi}^2 - \left\| \overline{D} \alpha \right\|_{\varphi}^2 \\ &\leq \left\| \overline{D}_{\varphi}^* \alpha \right\|_{\varphi}^2 \end{aligned}$$

Now we will see how Step 3 leads to the proof. We want to solve

$$\overline{D}u = g, \text{ where } g \text{ satisfies } \int_{\Omega} \frac{|g|^2}{\Delta \varphi} e^{-\varphi} dV = G^2 (< \infty).$$

We have the operator  $\overline{D}_{\varphi}^* : \mathcal{C}_c^2(\Omega) \rightarrow L^2(\Omega, e^{-\varphi})$  whose range is a much smaller set. Let  $\mathcal{R} = \text{Range}(\overline{D}_{\varphi}^*) \subset \mathcal{C}_c^1(\Omega) \subset L^2(\Omega, e^{-\varphi})$ . So there is a linear map  $L : \mathcal{R} \rightarrow \mathbb{C}$  given by

$$L(\overline{D}_{\varphi}^* \alpha) = \langle \alpha, g \rangle_{\varphi}.$$

Now we claim that  $L$  is a bounded linear functional on  $\mathcal{R}$ . This can be seen as follows:

$$\begin{aligned}
|L(\overline{D}_\varphi^* \alpha)| &= \left| \int_{\Omega} \alpha \bar{g} e^{-\varphi} dV \right| \\
&= \left| \int_{\Omega} \left( \sqrt{\Delta \varphi} \alpha \right) \left( \frac{\bar{g}}{\sqrt{\Delta \varphi}} \right) e^{-\varphi} dV \right| \quad (\text{since } \Delta \varphi \neq 0) \\
&\leq \left( \int_{\Omega} \Delta \varphi |\alpha|^2 e^{-\varphi} dV \right)^{1/2} \left( \int_{\Omega} \frac{|g|^2}{\Delta \varphi} e^{-\varphi} dV \right)^{1/2} \quad (\text{by Cauchy-Schwartz inequality}) \\
&\leq \left\| \overline{D}_\varphi^* \alpha \right\|_{\varphi} \cdot G \quad (\text{using apriori estimate})
\end{aligned}$$

Hence  $L$  is bounded on  $\mathcal{R} \subset L^2(\Omega, e^{-\varphi})$ . Now using Hahn-Banach extension theorem,  $L$  extends to a bounded linear functional (again denoted by  $L$ )  $L : L^2(\Omega, e^{-\varphi}) \rightarrow \mathbb{C}$  such that  $\|L\| \leq G$ .

Note: Since we are in a Hilbert space, there are easier ways of extending a bounded linear operator from a proper subspace to the bigger space (Declare the functional to be 0 on the orthogonal complement of the subspace).

By Reisz-Representation theorem, there is a  $u \in L^2(\Omega, e^{-\varphi})$  such that  $L(f) = \langle f, u \rangle_{\varphi}$ . Now if we choose  $f \in \mathcal{R}$ ,

$$L(\overline{D}_\varphi^* \alpha) = \left\langle \overline{D}_\varphi^* \alpha, u \right\rangle_{\varphi}.$$

But by definition,  $L(\overline{D}_\varphi^* \alpha) = \langle \alpha, g \rangle_{\varphi}$ . This gives us

$$\langle \alpha, g \rangle_{\varphi} = \left\langle \overline{D}_\varphi^* \alpha, u \right\rangle_{\varphi} = \langle \alpha, \overline{D}u \rangle_{\varphi}.$$

Since this is true for every  $\alpha \in \mathcal{C}_c^2(\Omega)$ , we get  $\overline{D}u = g$ .

Furthermore, by the Reisz-Representation theorem we know that  $\|u\| = \|L\| \leq G$ . ■